COMPARING TWO CONSTRUCTIONS OF THE LOCALIZATION OF A MORPHISM OF TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's argument, namely the comparison of the two descriptions of the induced pullback; the closing sentence 'Some details omitted' is preserved verbatim inside the counted span and concerns routine bookkeeping, not the main line of the proof.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The localization morphism attached to a sheaf. Same chapter, Lemma lemma-localize-morphism-topoi and Lemma lemma-localize-morphism. Supply the two diagrams of morphisms of topoi that are compared.
- The comparison of the localizations of a topos. Same chapter, Lemma lemma-localize-compare. Supplies the identification of the localizing sheaves under which the diagrams are compared.
- The pullback of a representable sheaf. Same chapter, Lemma lemma-pullback-representable-sheaf. Identifies the localizing sheaf with the pullback of the other one.

Source reference. Original lemma 04D3: sources/stacks-sites/pinned-original-source.tex, source lines 7875–7875. Navigation pointer invoked at proof line 7875; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04H1: sources/stacks-sites/pinned-original-source.tex, source lines 7904–7904. Navigation pointer invoked at proof line 7904; the counted span contains the pointer only and the cited result is not reproduced inside it, so

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Source reference. Original lemma 03CF: sources/stacks-sites/pinned-original-source.tex, source lines 7877–7877. Navigation pointer invoked at proof line 7877; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04IQ: sources/stacks-sites/pinned-original-source.tex, source lines 7888–7888. Navigation pointer invoked at proof line 7888; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : \mathcal{C} \to \mathcal{D}$ be a morphism of sites given by the continuous functor $u : \mathcal{D} \to \mathcal{C}$. Let V be an object of $\mathcal{D}$. Set $U = u(V)$. Set $\mathcal{G} = h_V^\#$, and $\mathcal{F} = h_U^\# = f^{-1}h_V^\#$ (see Lemma 04D3). Then the diagram of morphisms of topoi of Lemma 04H1 agrees with the diagram of morphisms of topoi of Lemma 03CF via the identifications $j_\mathcal{F} = j_U$ and $j_\mathcal{G} = j_V$ of Lemma 04IQ.*

Proof. This is not a complete triviality as the choice of morphism of sites giving rise to f made in the proof of Lemma 04H1 may be different from the morphisms of sites given to us in the lemma. But in both cases the functor $(f')^{-1}$ is described by the same rule. Hence they agree and the associated morphism of topoi is the same. Some details omitted. □